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Inventor(s)	Dou; Jinbo et al.

SENSOR MEMBRANES FOR DEXAMETHOSONE REJECTION

Abstract

Embodiments of the invention provide amperometric analyte sensors having optimized elements such as dexamethasone rejection membranes as well as methods for making and using such sensors. The amperometric analyte sensor apparatus comprises: a base layer; a conductive layer disposed on the base layer and comprising a working electrode; an dexamethasone rejection membrane disposed over an electroactive surface of the working electrode, wherein the interference rejection membrane comprises a poly Hema composition and an analyte sensing layer. While embodiments of the innovation can be used in a variety of contexts, typical embodiments of the invention include glucose sensors used in the management of diabetes.

Inventors: **Dou; Jinbo (Northridge, CA), Rao; Ashwin K. (West Hills, CA), Garai; Ellis (Woodland Hills, CA)**

Applicant: **Medtronic MiniMed, Inc.** (Northridge, CA)

Family ID: **94869318**

Assignee: **Medtronic MiniMed, Inc. (Northridge, CA)**

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Patent Application No. 63/562,394, filed Mar. 7, 2024, entitled “SENSOR MEMBRANES FOR DEXAMETHOSONE REJECTION”, which application is incorporated by reference herein.

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] Amperometric analyte sensors (e.g., glucose sensors used in the management of diabetes) and methods and materials for making and using such sensors.
2. Description of Related Art
[0003] Analyte sensors such as biosensors include devices that use biological elements to convert a chemical analyte in a matrix into a detectable signal. There are many types of biosensors used for a wide variety of analytes. The most studied type of biosensor is the amperometric glucose sensor, which is crucial to the successful glucose level control for diabetes.
[0004] A typical glucose sensor works according to the following chemical reactions:
##STR00001##
The glucose oxidase is used to catalyze the reaction between glucose and oxygen to yield gluconic acid and hydrogen peroxide (Equation 1). The H.sub.2O.sub.2 reacts electrochemically as shown in Equation 2, and the current can be measured by a potentiostat. These reactions, which occur in a variety of oxidoreductases known in the art, are used in a number of sensor designs.
[0005] Dexamethasone acetate (DXAC) can be used as an anti-inflammatory agent with implantable analyte sensors in order to, for example, extend sensor in vivo life-times. However, amperometric sensor reading (aka iSig) can be negatively impacted when exposed to DXAC. For example, certain continuous glucose monitoring (CGM) amperometric sensor designs experience a decrease in iSig when exposed to DXAC. In this context, minimizing the impact of DXAC on such sensors is desirable in order to maintain and improve sensor accuracy.
[0006] For the reasons noted above, methods and materials designed to address amperometric sensor difficulties caused by the use of dexamethasone are desirable.
SUMMARY OF THE INVENTION
[0007] Embodiments of the invention described herein include a dexamethasone acetate rejection membrane (DRM), made from materials selected to prevent dexamethasone from penetrating into sensing elements of amperometric sensors. In illustrative embodiments, this membrane can be composed of various poly(2-hydroxyethyl methacrylate) (“polyHema”) compositions in various formulations, and can be incorporated into the sensor in various locations. The DRM can consist of pre-polymerized polyHEMA or a polymerization reaction mixture (e.g. one comprising HEMA monomers) with no pre-polymerized polyHEMA molecules. Embodiments of the invention can further tailor the thickness and permeability of the DRM layer such that it does not interfere (or limit) the glucose diffusivity of the sensor while simultaneously preventing dexamethasone from impacting the glucose sensor signal. Moreover, the DRM can be a thin layer (like a skim coat) placed on top of the outermost chemistry layer of the glucose sensor, which is typically the glucose limiting membrane (GLM). Alternatively, the DRM can be placed under the GLM. In certain embodiments of the invention, there is an additional advantage in that the DRM layer can further reduce the impact of acetaminophen (AC) and other interferents on the glucose sensor signal, thus further improving sensor signal accuracy.

[0008] As noted above, the present invention provides methods and materials designed to maintain and improve sensor accuracy, and address sensor reading difficulties caused by dexamethasone. Embodiments of the invention include membrane compositions that can be used, for example, as barriers for dexamethasone in amperometric analyte sensors and sensor systems such as amperometric glucose sensors that are commonly used in the management of diabetes. In particular, embodiments of the invention include dexamethasone rejection membranes (DRMs). Sensors including these membranes show dexamethasone rejection, good oxygen effect character, and in vivo and in vitro performance.

[0009] The invention disclosed herein has a number of embodiments. Embodiments of the invention include, for example, methods of making a sensor apparatus for implantation within a mammal. Typically these methods comprise the steps of: providing a base layer; forming a conductive layer on the base layer, wherein the conductive layer includes a working electrode; forming a dexamethasone rejection membrane over the working electrode; forming an analyte sensing layer, wherein the analyte sensing layer includes an oxidoreductase; and forming an analyte modulating layer, wherein the analyte modulating layer includes a composition that modulates the diffusion of the analyte therethrough.

[0010] In certain embodiments of the methods of making a sensor apparatus for implantation within a mammal, the dexamethasone rejection membrane comprises about: 50%-80% (w/v) tetra (ethylene glycol) diacrylate; in combination with 0.5%-10% (w/v) methacryloyloxyethyl phosphorylcholine (MPC). In some embodiments of the methods of making a sensor apparatus for implantation within a mammal, the dexamethasone rejection membrane comprises about: 60%-70% (w/v) tetra (ethylene glycol) diacrylate; in combination with 1.0%-1% (w/v) methacryloyloxyethyl phosphorylcholine (MPC).

[0011] Embodiments of the invention also include sensor apparatus for implantation within a mammal. Typically such sensors include: a base layer; a conductive layer disposed on the base layer, wherein the conductive layer includes a working electrode; an dexamethasone rejection membrane disposed over the working electrode, wherein the dexamethasone rejection membrane comprises a poly(2-hydroxyethyl methacrylate) composition; an analyte sensing layer disposed over the dexamethasone rejection membrane, wherein the analyte sensing layer includes an oxidoreductase; and an analyte modulating layer disposed over the analyte sensing layer, wherein the analyte modulating layer includes a composition that modulates the diffusion of the analyte therethrough.

[0012] Other embodiments of the invention include methods of estimating the concentrations of an analyte such as glucose in vivo, these methods comprising disposing a sensor apparatus disclosed herein into an in vivo environment of a subject, wherein the environment comprises the analyte such as glucose; and estimating the concentration of analyte; so that the concentrations of analyte in vivo are estimated.

[0013] Other objects, features and advantages of the present invention will become apparent to those skilled in the art from the following detailed description. It is to be understood, however, that the detailed description and specific examples, while indicating some embodiments of the present invention are given by way of illustration and not limitation. Many changes and modifications within the scope of the present invention may be made without departing from the spirit thereof, and the invention includes all such modifications.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0014] FIG. **1** provides a schematic of the well known reaction between glucose and glucose oxidase. As shown in a stepwise manner, this reaction involves glucose oxidase (GOx), glucose, and oxygen in water. In the reductive half of the reaction, two protons and electrons are transferred from β-D-glucose to the enzyme yielding d-gluconolactone. In the oxidative half of the reaction, the enzyme is oxidized by molecular oxygen yielding hydrogen peroxide. The d-gluconolactone then reacts with water to hydrolyze the lactone ring and produce gluconic acid. In certain electrochemical sensors of the invention, the hydrogen peroxide produced by this reaction is oxidized at the working electrode (H.sub.2O.sub.2.fwdarw.2H.sup.++O.sub.2+2e.sup.−).

[0015] FIG. **2**A provides a diagrammatic view of an embodiment of an amperometric analyte sensor to which a dexamethasone rejection membrane can be added. FIG. **2**B provides a diagrammatic view of an embodiment of an amperometric analyte sensor having a dexamethasone rejection membrane **120**.

[0016] FIGS. **3**A-**3**B. FIG. **3**A provides three panels showing diagrammatic views of embodiments of an amperometric analyte sensor having dexamethasone rejection membranes. The upper left and right panels provide diagrammatic view of embodiments of amperometric analyte sensors comprising a plurality of layered elements including a dexamethasone rejection membrane. The lower panel provides a cartoon schematic of methods and molecules useful to make PHEMA membranes. FIG. **3**B provides three panels showing diagrammatic views of embodiments of an amperometric analyte sensor having dexamethasone rejection membranes. The left panel provides a diagrammatic view of embodiments of amperometric analyte sensor comprising a plurality of layered elements including a dexamethasone rejection membrane disposed within the layers. The middle panel provides a diagrammatic view of embodiments of amperometric analyte sensor comprising a plurality of layered elements including a dexamethasone rejection membrane disposed as an external layer. The right panel provides a diagrammatic view of embodiments of amperometric analyte sensor comprising a plurality of layered elements including a dexamethasone rejection membrane disposed as an external layer as well as a dexamethasone rejection membrane disposed within the layers.

[0017] FIG. **4** provides schematics in table form of various dexamethasone rejection membrane formulations. The top panel includes illustrative dexamethasone rejection membrane components and their ratios. The bottom panel includes illustrative dexamethasone rejection membrane components having different permeability profiles.

[0018] FIG. **5** provides data from studies of sensors having a DRM as compared to control sensors that do not include a DRM. This graph provides data from normalized iSig sensor readings in these two sensor embodiments in the presence of dexamethasone. This data shows that sensors comprising a DRM exhibited less fluctuations in iSig (nA) following exposure to DXAC; as well as improved stability (i.e., less change is glucose response (day 2 vs. day 4) after acetaminophen (AC) and DXAC exposure. The data in the upper left graph shows that sensors comprising a DRM exhibited improved/faster run in times.

[0019] FIG. **6** provides data from sensitivity analysis studies (nA/mg/dl) of sensors having DRMs made from different formulations as compared to control sensors that do not include a DRM. As shown for example by the boxed datapoints in the right side of the graphs, sensors designed to include a poly(2-hydroxyethyl methacrylate) composition including 50-80% (w/v) tetra (ethylene glycol) diacrylate; in combination with 0%-10% (w/v) methacryloyloxyethyl phosphorylcholine (MPC) is more stable (smaller change in sensitivity from Day 2 to Day 4) compared to control sensors not comprising a DRM.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Unless otherwise defined, all terms of art, notations, and other scientific terms or terminology used herein are intended to have the meanings commonly understood by those of skill in the art to which this invention pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art. Many of the techniques and procedures described or referenced herein are well understood and commonly employed using conventional methodology by those skilled in the art. As appropriate, procedures involving the use of commercially available kits and reagents are generally carried out in accordance with manufacturer defined protocols and/or parameters unless otherwise noted. In addition, certain text from related art is reproduced herein to more clearly delineate the various embodiments of the invention. A number of terms are defined below.

[0021] All publications mentioned herein are expressly incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. Publications cited herein are cited for their disclosure prior to the filing date of the present application. Nothing here is to be construed as an admission that the inventors are not entitled to antedate the publications by virtue of an earlier priority date or prior date of invention. Further the actual publication dates may be different from those shown and require independent verification.

[0022] It must be noted that as used herein and in the appended claims, the singular forms “a”, “and”, and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an oxidoreductase” includes a plurality of such oxidoreductases and equivalents thereof known to those skilled in the art, and so forth. All numbers recited in the specification and associated claims that refer to values that can be numerically characterized with a value other than a whole number (e.g. the concentration of a compound in a solution) are understood to be modified by the term “about”.

[0023] The term “oxidoreductase” is used according to its art accepted meaning, i.e. an enzyme that catalyzes the transfer of electrons from one molecule (the reductant, also called the hydrogen or electron donor) to another (the oxidant, also called the hydrogen or electron acceptor). Typical oxidoreductases include glucose oxidase and lactate oxidase. The term “carrier polypeptide” or “carrier protein” is used according to its art accepted meaning of an additive included to maintain the stability of a polypeptide, for example the ability of an oxidoreductase polypeptide to maintain certain qualitative features such as physical and chemical properties (e.g. an ability to oxidize glucose) of a composition comprising a polypeptide for a period of time. A typical carrier protein commonly used in the art is albumin.

[0024] The term “analyte” as used herein is a broad term and is used in its ordinary sense, including, without limitation, to refer to a substance or chemical constituent in a fluid such as a biological fluid (for example, blood, interstitial fluid, cerebral spinal fluid, lymph fluid or urine) that can be analyzed. Analytes can include naturally occurring substances, artificial substances, metabolites, and/or reaction products. In some embodiments, the analyte for measurement by the sensing regions, devices, and methods is glucose. However, other analytes are contemplated as well, including but not limited to, lactate. Salts, sugars, proteins fats, vitamins and hormones naturally occurring in blood or interstitial fluids can constitute analytes in certain embodiments. The analyte can be naturally present in the biological fluid or endogenous; for example, a metabolic product, a hormone, an antigen, an antibody, and the like. Alternatively, the analyte can be introduced into the body or exogenous, for example, a contrast agent for imaging, a radioisotope, a chemical agent, a fluorocarbon-based synthetic blood, or a drug or pharmaceutical composition, including but not limited to insulin. The metabolic products of drugs and pharmaceutical compositions are also contemplated analytes.

[0025] The term “sensor,” as used herein, is a broad term and is used in its ordinary sense, including, without limitation, the portion or portions of an analyte-monitoring device that detects an analyte. In one embodiment, the sensor includes an electrochemical cell that has a working electrode, a reference electrode, and optionally a counter electrode passing through and secured within the sensor body forming an electrochemically reactive surface at one location on the body, an electronic connection at another location on the body, and a membrane system affixed to the body and covering the electrochemically reactive surface. During general operation of the sensor, a biological sample (for example, blood or interstitial fluid), or a portion thereof, contacts (directly or

after passage through one or more membranes or domains) an enzyme (for example, glucose oxidase); the reaction of the biological sample (or portion thereof) results in the formation of reaction products that allow a determination of the analyte level in the biological sample.
[0026] The terms “electrochemically reactive surface” and “electroactive surface” as used herein are broad terms and are used in their ordinary sense, including, without limitation, the surface of an electrode where an electrochemical reaction takes place. In one example, a working electrode (e.g. one comprised of platinum black) measures the hydrogen peroxide produced by the enzyme-catalyzed reaction of the analyte being detected by creating an electric current. For example, detection of glucose analyte utilizing glucose oxidase produces H.sub.2O.sub.2 as a by-product. The H.sub.2O.sub.2 reacts with the surface of the working electrode producing two protons (2H.sup.+), two electrons (2e.sup.−), and one molecule of oxygen (O.sub.2) which produces the electronic current being detected. In the case of the counter electrode, a reducible species, for example, O.sub.2 is reduced at the electrode surface in order to balance the current being generated by the working electrode.
[0027] The term “sensing region” as used herein is a broad term and is used in its ordinary sense, including, without limitation, the region of a monitoring device responsible for the detection of a particular analyte. In an illustrative embodiment, the sensing region can comprise a non-conductive body, a working electrode, a reference electrode, and a counter electrode passing through and secured within the body forming electrochemically reactive surfaces on the body and an electronic connective means at another location on the body, and a one or more layers covering the electrochemically reactive surface.
[0028] As discussed in detail below, embodiments of the invention relate to the use of an electrochemical sensor that exhibits a novel constellation of elements including an dexamethasone rejection membrane selected to have a unique set of technically desirable material properties. The electrochemical sensors of the invention are designed to be implanted to measure a concentration of an analyte of interest (e.g. glucose) or a substance indicative of the concentration or presence of the analyte in fluid. Typically, these sensor include an agent such as dexamethasone selected to inhibit or ameliorate immune or inflammatory responses. In some embodiments, the sensor is a continuous device, for example a subcutaneous, transdermal, or intravascular device. In some embodiments, the device can analyze a plurality of intermittent blood samples. The sensor embodiments disclosed herein can use any known method, including invasive, minimally invasive, and non-invasive sensing techniques, to provide an output signal indicative of the concentration of the analyte of interest. Typically, the sensor is of the type that senses a product or reactant of an enzymatic reaction between an analyte and an enzyme in the presence of oxygen as a measure of the analyte in vivo or in vitro. Such sensors typically comprise a membrane surrounding the enzyme through which an analyte migrates. The product is then measured using electrochemical methods and thus the output of an electrode system functions as a measure of the analyte. In some embodiments, the sensor can use an amperometric, coulometric, conductimetric, and/or potentiometric technique for measuring the analyte.
[0029] Embodiments of the invention disclosed herein provide sensors of the type used, for example, in subcutaneous or transcutaneous monitoring of blood glucose levels in a diabetic patient. A variety of implantable, electrochemical biosensors have been developed for the treatment of diabetes and other life-threatening diseases. Many existing sensor designs use some form of immobilized enzyme to achieve their bio-specificity. Embodiments of the invention described herein can be adapted and implemented with a wide variety of known electrochemical sensors, including for example, U.S. Patent Application No. 20050115832, U.S. Pat. Nos. 6,001,067, 6,702,857, 6,212,416, 6,119,028, 6,400,974, 6,595,919, 6,141,573, 6,122,536, 6,512,939 5,605,152, 4,431,004, 4,703,756, 6,514,718, 5,985,129, 5,390,691, 5,391, 250, 5,482,473, 5,299,571, 5,568,806, 5,494,562, 6,120,676, 6,542,765 as well as PCT International Publication Numbers WO 01/58348, WO 04/021877, WO 03/034902, WO 03/035117, WO 03/035891, WO

03/023388, WO 03/022128, WO 03/022352, WO 03/023708, WO 03/036255, WO03/036310 WO 08/042625, and WO 03/074107, and European Patent Application EP 1153571, the contents of each of which are incorporated herein by reference.

[0030] As discussed in detail below, embodiments of the invention disclosed herein provide sensor elements having enhanced material properties and/or architectural configurations and sensor systems (e.g. those comprising a sensor and associated electronic components such as a monitor, a processor and the like) constructed to include such elements. The disclosure further provides methods for making and using such sensors and/or architectural configurations. While some embodiments of the invention pertain to ketone (see, e.g., U.S. patent application Ser. No. 17/501,292, the contents of which are incorporated by reference) glucose (see, e.g., U.S. patent application Ser. No. 15/140,129, the contents of which are incorporated by reference), a variety of the elements disclosed herein (e.g., dexamethasone rejection membranes) can be adapted for use with any one of the wide variety of sensors known in the art. The analyte sensor elements, architectures, and methods for making and using these elements that are disclosed herein can be used to establish a variety of layered sensor structures. Such sensors of the invention exhibit a surprising degree of flexibility and versatility, characteristics which allow a wide variety of sensor configurations to be designed to examine a wide variety of analyte species.

[0031] Specific aspects of embodiments of the invention are discussed in detail in the following sections.

I. Typical Elements, Configurations and Analyte Sensor Embodiments of the Invention

[0032] A wide variety of sensors and sensor elements are known in the art including amperometric sensors used to detect and/or measure biological analytes such as glucose. Many glucose sensors are based on an oxygen (Clark-type) amperometric transducer (see, e.g. Yang et al., Electroanalysis 1997, 9, No. 16: 1252-1256; Clark et al., Ann. N.Y. Acad. Sci. 1962, 102, 29; Updike et al., Nature 1967, 214,986; and Wilkins et al., Med. Engin. Physics, 1996, 18, 273.3-51). A number of in vivo glucose sensors utilize amperometric sensors because such transducers are relatively easy to fabricate and can readily be miniaturized using conventional technology. A problem associated with the use of implantable amperometric sensors that include dexamethasone, however, is signal interference. As discussed in detail below, these problems are addressed by the novel membranes disclosed herein that modulate the transport properties of dexamethasone that can create a signal in amperometric sensors. Consequently, these membranes can be used with any of a variety of amperometric sensors used in analyte sensors that are susceptible to interference from dexamethasone.

[0033] Amperometric sensors typically comprise a plurality of layered elements including, for example, a base layer having an electrode, an enzyme layer, and an analyte diffusion control (e.g. glucose limiting) membrane. In some sensor embodiments, an adhesion promoter layer is added to facilitate close attachment of various layers such as a diffusion control membrane and enzyme layer. One such sensor embodiment is shown in FIG. **2**A. Embodiments of the invention disclosed herein can further include a dexamethasone rejection membrane (DRM) that is designed to inhibit and/or prevent dexamethasone from accessing the sensor electrode and confounding measurements of the signal generated by the analyte to be measured. Illustrative embodiments of a sensor having a dexamethasone rejection membrane are shown in FIG. **3-4**.

[0034] The dexamethasone rejection membrane compositions and associated processes disclosed herein significantly improve sensor reliability. Embodiments of the dexamethasone rejection membranes disclosed herein comprise materials constructed to exhibit a constellation of material properties that can be used in a variety of contexts and for example, overcome a number of technical problems observed in sensors, such as electrochemical glucose sensors that are implanted in vivo and which utilize the chemical reaction between glucose and glucose oxidase to generate a measurable signal. Specifically, various embodiments of a dexamethasone rejecting membrane (DRM) is provided that retards the diffusion of interfering dexamethasone molecules.

Embodiments of the invention include membrane compositions that can be used for example in amperometric sensors (e.g. glucose sensors used by individuals suffering from diabetes) to inhibit spurious signals caused by dexamethasone.

[0035] In addition to the above-noted material properties, in various embodiments, the molecular structure of the dexamethasone rejection membrane material is free of electrochemically reactive moieties that can for example, generate spurious signals at the surface of an electrode (either directly or indirectly). Moreover, for sensors that comprise a plurality of layered elements, in certain embodiments, the material of the dexamethasone rejection layer exhibits adhesive properties that allow it to adhere to coarse surfaces such as the platinum black compositions that comprise some electrode surfaces, while at the same time allow it to adhere to a variety of other matrices, for example adjacent layers comprising bioactive molecules such as glucose oxidase (i.e. so as to exhibit adhesive properties that inhibit delamination of the sensor layers).

[0036] In addition, sensors comprising a plurality of layers that are designed to measure analytes in aqueous environments (e.g., those implanted in vivo) typically require wetting of the layers prior to and during the measurement of accurate analyte reading. Because the properties of a material can influence the rate at which it hydrates, in various embodiments, the material properties of the dexamethasone rejection membrane used in aqueous environments facilitates sensor wetting (“run in”) to, for example, minimize the time period between the sensor's introduction into an aqueous environment and its ability to provide accurate signals that correspond to the concentrations of an analyte in that environment.

[0037] With electrochemical glucose sensors that utilize the chemical reaction between glucose and glucose oxidase to generate a measurable signal, various embodiments of the dexamethasone rejection membrane do not exacerbate (and in some instances, actually diminish) what is known in the art as the “oxygen deficit problem”. Specifically, because glucose oxidase-based sensors require both oxygen (O.sub.2) as well as glucose to generate a signal, the presence of an excess of oxygen relative to glucose, is necessary for the operation of a glucose oxidase-based glucose sensor. However, because the concentration of oxygen in subcutaneous tissue is much less than that of glucose, oxygen can be the limiting reactant in the reaction between glucose, oxygen, and glucose oxidase in a sensor—a situation which compromises the sensor's ability to produce a signal that is strictly dependent on the concentration of glucose. In this context, because the properties of a DRM material can influence the rate at which compounds diffuse through that material to the site of a measurable chemical reaction, the material properties of the dexamethasone rejection membrane used in electrochemical glucose sensors that utilize the chemical reaction between glucose and glucose oxidase to generate a measurable signal, does not for example, favor the diffusion of glucose over oxygen in a manner that contributes to the oxygen deficit problem.

A. Typical Sensor Architectures Found in Embodiments of the Invention

[0038] FIG. **2**A illustrates a cross-section of a typical sensor embodiment **100** of the present invention. This sensor embodiment is formed from a plurality of components that are typically in the form of layers of various conductive and non-conductive constituents disposed on each other according to art accepted methods and/or the specific methods of the invention disclosed herein. The components of the sensor are typically characterized herein as layers because, for example, it allows for a facile characterization of the sensor structure shown in FIG. **2**. Artisans will understand however, that in certain embodiments of the invention, the sensor constituents are combined such that multiple constituents form one or more heterogeneous layers. In this context, those of skill in the art understand that the ordering of the layered constituents can be altered in various embodiments of the invention.

[0039] The embodiment shown in FIG. **2**A can include a base layer **102** to support the sensor **100** (e.g. disposed under the conductive layer **104**). The base layer **102** can be made of a material such as a metal and/or a ceramic and/or a polymeric substrate, which may be self-supporting or further supported by another material as is known in the art. Embodiments of the invention include a conductive layer **104** which is disposed on and/or combined with the base layer **102**. Typically the conductive layer **104** comprises one or more electrodes. An operating sensor **100** typically includes a plurality of electrodes such as a working electrode, a counter electrode and a reference electrode. Other embodiments may also include a plurality of working and/or counter and/or reference electrodes and/or one or more electrodes that performs multiple functions, for example one that functions as both as a reference and a counter electrode.

[0040] As discussed in detail below, the base layer **102** and/or conductive layer **104** can be generated using many known techniques and materials. In certain embodiments of the invention, the electrical circuit of the sensor is defined by etching the disposed conductive layer **104** into a desired pattern of conductive paths. A typical electrical circuit for the sensor **100** comprises two or more adjacent conductive paths with regions at a proximal end to form contact pads and regions at a distal end to form sensor electrodes. An electrically insulating cover layer **106** such as a polymer coating can be disposed on portions of the sensor **100**. Various sensor stack layers are contemplated. For example, in FIG. **2**A, perhaps insulating layer can be disposed over layer **104** (conductive layer). Acceptable polymer coatings for use as the insulating protective cover layer **106** can include, but are not limited to, non-toxic biocompatible polymers such as silicone compounds, polyimides, biocompatible solder masks, epoxy acrylate copolymers, or the like. In the sensors of the present invention, one or more exposed regions or apertures **108** can be made through the cover layer **106** to open the conductive layer **104** to the external environment and to, for example, allow an analyte such as glucose to permeate the layers of the sensor and be sensed by the sensing elements. Apertures **108** can be formed by a number of techniques, including laser ablation, tape masking, chemical milling or etching or photolithographic development or the like. In certain embodiments of the invention, during manufacture, a secondary photoresist can also be applied to the protective layer **106** to define the regions of the protective layer to be removed to form the aperture(s) **108**. The exposed electrodes and/or contact pads can also undergo secondary processing (e.g. through the apertures **108**), such as additional plating processing, to prepare the surfaces and/or strengthen the conductive regions.

[0041] In the sensor configuration shown in FIG. **2**A, an analyte sensing layer **110** (which is typically a sensor chemistry layer, meaning that materials in this layer undergo a chemical reaction to produce a signal that can be sensed by the conductive layer) is disposed on one or more of the exposed electrodes of the conductive layer **104**. In the sensor configuration shown in FIG. **2**B, an interference rejection membrane **120** is disposed on one or more of the exposed electrodes of the conductive layer **104**, with the analyte sensing layer **110** then being disposed on this interference rejection membrane **120**. Typically, the analyte sensing layer **110** is an enzyme layer. Most typically, the analyte sensing layer **110** comprises an enzyme capable of producing and/or utilizing oxygen and/or hydrogen peroxide, for example the enzyme glucose oxidase. Optionally, the enzyme in the analyte sensing layer is combined with a second carrier protein such as human serum albumin, bovine serum albumin or the like. In an illustrative embodiment, an oxidoreductase enzyme such as glucose oxidase in the analyte sensing layer **110** reacts with glucose to produce hydrogen peroxide, a compound which then modulates a current at an electrode. As this modulation of current depends on the concentration of hydrogen peroxide, and the concentration of hydrogen peroxide correlates to the concentration of glucose, the concentration of glucose can be determined by monitoring this modulation in the current. In a specific embodiment of the invention, the hydrogen peroxide is oxidized at a working electrode which is an anode (also termed herein the anodic working electrode), with the resulting current being proportional to the hydrogen peroxide concentration. Such modulations in the current caused by changing hydrogen peroxide concentrations can by monitored by any one of a variety of sensor detector apparatuses such as a universal sensor amperometric biosensor detector or one of the other variety of similar devices known in the art such as glucose monitoring devices produced by Medtronic™ MiniMed™.

[0042] In embodiments of the invention, the analyte sensing layer **110** can be applied over portions of the conductive layer or over the entire region of the conductive layer. Typically the analyte sensing layer **110** is disposed on the working electrode which can be the anode or the cathode. Optionally, the analyte sensing layer **110** is also disposed on a counter and/or reference electrode. While the analyte sensing layer **110** can be up to about 1000 microns (μm) in thickness, the analyte sensing layer may be relatively thin as compared to those found in sensors previously described in the art, and is for example, typically less than 1, 0.5, 0.25 or 0.1 microns in thickness. In other instances, the analyte sensing layer **110** is of a thickness between 3.5 to 10 microns in thickness. As discussed in detail below, some methods for generating a thin analyte sensing layer **110** include brushing the layer onto a substrate (e.g. the reactive surface of a platinum black electrode), as well as spin coating processes, slot coating processes, dip and dry processes, low shear spraying processes, ink-jet printing processes, silk screen processes and the like.

[0043] Typically, the analyte sensing layer **110** is coated and/or disposed next to one or more additional layers. Optionally, the one or more additional layers includes a protein layer **116** disposed upon the analyte sensing layer **110**. Typically, the protein layer **116** comprises a protein such as human serum albumin, bovine serum albumin or the like. Typically, the protein layer **116** comprises human serum albumin. In some embodiments of the invention, an additional layer includes an analyte modulating layer **112** that is disposed above the analyte sensing layer **110** to regulate analyte access with the analyte sensing layer **110**. For example, the analyte modulating membrane layer **112** can comprise a glucose limiting membrane, which regulates the amount of glucose that contacts an enzyme such as glucose oxidase that is present in the analyte sensing layer. Such glucose limiting membranes can be made from a wide variety of materials known to be suitable for such purposes, e.g., silicone compounds such as polydimethyl siloxanes, polyurethanes, polyurea cellulose acetates, NAFION, polyester sulfonic acids (e.g. Kodak AQ), hydrogels or any other suitable hydrophilic membranes known to those skilled in the art.

[0044] In typical embodiments of the invention, an adhesion promoter layer **114** is disposed between the analyte modulating layer **112** and the analyte sensing layer **110** as shown in FIG. **2** in order to facilitate their contact and/or adhesion. In a specific embodiment of the invention, an adhesion promoter layer **114** is disposed between the analyte modulating layer **112** and the protein layer **116** as shown in FIG. **2** in order to facilitate their contact and/or adhesion. The adhesion promoter layer **114** can be made from any one of a wide variety of materials known in the art to facilitate the bonding between such layers. Typically, the adhesion promoter layer **114** comprises a silane compound. In alternative embodiments, protein or like molecules in the analyte sensing layer **110** can be sufficiently crosslinked or otherwise prepared to allow the analyte modulating membrane layer **112** to be disposed in direct contact with the analyte sensing layer **110** in the absence of an adhesion promoter layer **114**.

[0045] Embodiments of typical elements used to make the sensors disclosed herein are discussed below.

B. Typical Analyte Sensor Constituents Used in Embodiments of the Invention

[0046] The following disclosure provides examples of typical elements/constituents used in sensor embodiments of the invention. While these elements can be described as discreet units (e.g. layers), those of skill in the art understand that sensors can be designed to contain elements having a combination of some or all of the material properties and/or functions of the elements/constituents discussed below (e.g. an element that serves both as a supporting base constituent and/or a conductive constituent and/or a matrix for the analyte sensing constituent and which further functions as an electrode in the sensor). Those in the art understand that these thin film analyte sensors can be adapted for use in a number of sensor systems such as those described herein.

Base Constituent

[0047] Sensors of the invention typically include a base constituent (see, e.g. element **102** in FIG. **2**A). The term "base constituent" is used herein according to art accepted terminology and refers to the constituent in the apparatus that typically provides a supporting matrix for the plurality of constituents that are stacked on top of one another and comprise the functioning sensor. In one form, the base constituent comprises a thin film sheet of insulative (e.g. electrically insulative and/or water impermeable) material. This base constituent can be made of a wide variety of materials having desirable qualities such as dielectric properties, water impermeability and hermeticity. Some materials include metallic, and/or ceramic and/or polymeric substrates or the like.

[0048] The base constituent may be self-supporting or further supported by another material as is known in the art. In one embodiment of the sensor configuration shown in FIG. **2**A, the base constituent **102** comprises a ceramic. Alternatively, the base constituent comprises a polymeric material such as a polyimide. In an illustrative embodiment, the ceramic base comprises a composition that is predominantly Al.sub.2O.sub.3 (e.g. 96%). The use of alumina as an insulating base constituent for use with implantable devices is disclosed in U.S. Pat. Nos. 4,940,858, 4,678,868 and 6,472,122 which are incorporated herein by reference. The base constituents of the invention can further include other elements known in the art, for example hermetical vias (see, e.g. WO 03/023388). Depending upon the specific sensor design, the base constituent can be relatively thick constituent (e.g. thicker than 50, 100, 200, 300, 400, 500 or 1000 microns). Alternatively, one can utilize a nonconductive ceramic, such as alumina, in thin constituents, e.g., less than about 30 microns.

Conductive Constituent

[0049] The electrochemical sensors of the invention typically include a conductive constituent disposed upon the base constituent that includes at least one electrode for measuring an analyte or its byproduct (e.g. oxygen and/or hydrogen peroxide) to be assayed (see, e.g. element **104** in FIG. **2**A). The term “conductive constituent” is used herein according to art accepted terminology and refers to electrically conductive sensor elements such as electrodes which are capable of measuring a detectable signal and conducting this to a detection apparatus. An illustrative example of this is a conductive constituent that can measure an increase or decrease in current in response to exposure to a stimuli such as the change in the concentration of an analyte or its byproduct as compared to a reference electrode that does not experience the change in the concentration of the analyte, a coreactant (e.g. oxygen) used when the analyte interacts with a composition (e.g. the enzyme glucose oxidase) present in analyte sensing constituent **110** or a reaction product of this interaction (e.g. hydrogen peroxide). Illustrative examples of such elements include electrodes which are capable of producing variable detectable signals in the presence of variable concentrations of molecules such as hydrogen peroxide or oxygen. Typically one of these electrodes in the conductive constituent is a working electrode, which can be made from non-corroding metal or carbon. A carbon working electrode may be vitreous or graphitic and can be made from a solid or a paste. A metallic working electrode may be made from platinum group metals, including palladium or gold, or a non-corroding metallically conducting oxide, such as ruthenium dioxide. Alternatively the electrode may comprise a silver/silver chloride electrode composition. The working electrode may be a wire or a thin conducting film applied to a substrate, for example, by coating or printing. Typically, only a portion of the surface of the metallic or carbon conductor is in electrolytic contact with the analyte-containing solution. This portion is called the working surface of the electrode. The remaining surface of the electrode is typically isolated from the solution by an electrically insulating cover constituent **106**. Examples of useful materials for generating this protective cover constituent **106** include polymers such as polyimides, polytetrafluoroethylene, polyhexafluoropropylene and silicones such as polysiloxanes.

[0050] In addition to the working electrode, the analyte sensors of the invention typically include a reference electrode or a combined reference and counter electrode (also termed a quasi-reference electrode or a counter/reference electrode). If the sensor does not have a counter/reference electrode then it may include a separate counter electrode, which may be made from the same or different materials as the working electrode. Typical sensors of the present invention have one or more working electrodes and one or more counter, reference, and/or counter/reference electrodes. One embodiment of the sensor of the present invention has two, three or four or more working electrodes. These working electrodes in the sensor may be integrally connected or they may be kept separate.

[0051] Typically for in vivo use, embodiments of the present invention are implanted subcutaneously in the skin of a mammal for direct contact with the body fluids of the mammal, such as blood. Alternatively the sensors can be implanted into other regions within the body of a mammal such as in the intraperitoneal or subcutaneous space. When multiple working electrodes are used, they may be implanted together or at different positions in the body. The counter, reference, and/or counter/reference electrodes may also be implanted either proximate to the working electrode(s) or at other positions within the body of the mammal. Embodiments of the invention include sensors comprising electrodes constructed from nanostructured materials. As used herein, a “nanostructured material” is an object manufactured to have at least one dimension smaller than 100 nm. Examples include, but are not limited to, single-walled nanotubes, double-walled nanotubes, multi-walled nanotubes, bundles of nanotubes, fullerenes, cocoons, nanowires, nanofibres, onions and the like.

Dexamethasone Rejection Constituent

[0052] The electrochemical sensors of the invention typically include a dexamethasone rejection constituent disposed between the surface of the electrode and the environment to be assayed. Embodiments of the invention described herein include a dexamethasone acetate rejection membrane (DRM), made from materials selected to prevent dexamethasone from penetrating into sensing elements of amperometric sensors. In illustrative embodiments, this membrane can be composed of various poly(2-hydroxyethyl methacrylate) (“polyHema”) compositions in various forms, and can be incorporated into the sensor in various locations. The DRM can consist of pre-polymerized polyHEMA or no pre-polymerized polyHEMA. Embodiments of the invention tailor the thickness and permeability of the DRM layer such that it does not interfere (or limit) the analyte diffusivity of the sensor while simultaneously preventing dexamethasone from impacting the sensor signal.

Interference Rejection Constituent

[0053] The electrochemical sensors of the invention can include an interference rejection constituent disposed between the surface of the electrode and the environment to be assayed. In particular, certain sensor embodiments rely on the oxidation and/or reduction of hydrogen peroxide generated by enzymatic reactions on the surface of a working electrode at a constant potential applied. Because amperometric detection based on direct oxidation of hydrogen peroxide requires a relatively high oxidation potential, sensors employing this detection scheme may suffer interference from oxidizable species that are present in biological fluids, such as ascorbic acid, uric acid and acetaminophen. In this context, the term “interference rejection constituent” is used herein according to art accepted terminology and refers to a coating or membrane in the sensor that functions to inhibit spurious signals generated by such oxidizable species which interfere with the detection of the signal generated by the analyte to be sensed. Certain interference rejection constituents function via size exclusion (e.g. by excluding interfering species of a specific size).

[0054] Additional compositions having an unexpected constellation of material properties that make them ideal for use as interference rejection membranes in certain amperometric glucose sensors as well as methods for making and using them are further disclosed herein.

Analyte Sensing Constituent

[0055] The electrochemical sensors of the invention include an analyte sensing constituent disposed on the electrodes of the sensor (see, e.g. element **110** in FIG. **2**A). The term “analyte sensing constituent” is used herein according to art accepted terminology and refers to a constituent comprising a material that is capable of recognizing or reacting with an analyte whose presence is to be detected by the analyte sensor apparatus. Typically this material in the analyte sensing constituent produces a detectable signal after interacting with the analyte to be sensed, typically via the electrodes of the conductive constituent. In this regard the analyte sensing constituent and the electrodes of the conductive constituent work in combination to produce the electrical signal that is read by an apparatus associated with the analyte sensor. Typically, the analyte sensing constituent comprises an oxidoreductase enzyme capable of reacting with and/or producing a molecule whose change in concentration can be measured by measuring the change in the current at an electrode of the conductive constituent (e.g. oxygen and/or hydrogen peroxide), for example the enzyme glucose oxidase. An enzyme capable of producing a molecule such as hydrogen peroxide can be disposed on the electrodes according to a number of processes known in the art. The analyte sensing constituent can coat all or a portion of the various electrodes of the sensor. In this context, the analyte sensing constituent may coat the electrodes to an equivalent degree. Alternatively the analyte sensing constituent may coat different electrodes to different degrees, with, for example, the coated surface of the working electrode being larger than the coated surface of the counter and/or reference electrode.

[0056] Typical sensor embodiments of this element of the invention utilize an enzyme (e.g. glucose oxidase) that has been combined with a second protein (e.g. albumin) in a fixed ratio (e.g. one that is typically optimized for glucose oxidase stabilizing properties) and then applied on the surface of an electrode to form a thin enzyme constituent. In a typical embodiment, the analyte sensing constituent comprises a GOx and HSA mixture. In a typical embodiment of an analyte sensing constituent having GOx, the GOx reacts with glucose present in the sensing environment (e.g. the body of a mammal) and generates hydrogen peroxide according to the reaction shown in FIG. **1**, wherein the hydrogen peroxide so generated is anodically detected at the working electrode in the conductive constituent.

[0057] As noted above, the enzyme and the second protein (e.g. an albumin) are typically treated to form a crosslinked matrix (e.g. by adding a cross-linking agent to the protein mixture). As is known in the art, crosslinking conditions may be manipulated to modulate factors such as the retained biological activity of the enzyme, its mechanical and/or operational stability. Illustrative crosslinking procedures are described in U.S. patent application Ser. No. 10/335,506 and PCT publication WO 03/035891 which are incorporated herein by reference. For example, an amine cross-linking reagent, such as, but not limited to, glutaraldehyde, can be added to the protein mixture.

Protein Constituent

[0058] The electrochemical sensors of the invention optionally include a protein constituent disposed between the analyte sensing constituent and the analyte modulating constituent (see, e.g. element **116** in FIG. **2**A). The term “protein constituent” is used herein according to art accepted terminology and refers to a constituent containing a carrier protein or the like that is selected for compatibility with the analyte sensing constituent and/or the analyte modulating constituent. In typical embodiments, the protein constituent comprises an albumin such as human serum albumin. The HSA concentration may vary between about 0.5%-30% (w/v). Typically, the HSA concentration is about 1-10% w/v, and most typically is about 5% w/v. In alternative embodiments of the invention, collagen or BSA or other structural proteins used in these contexts can be used instead of or in addition to HSA. This constituent is typically crosslinked on the analyte sensing constituent according to art accepted protocols.

Adhesion Promoting Constituent

[0059] The electrochemical sensors of the invention can include one or more adhesion promoting (AP) constituents (see, e.g. element **114** in FIG. **2**A). The term “adhesion promoting constituent” is used herein according to art accepted terminology and refers to a constituent that includes materials selected for their ability to promote adhesion between adjoining constituents in the sensor. Typically, the adhesion promoting constituent is disposed between the analyte sensing constituent and the analyte modulating constituent. Typically, the adhesion promoting constituent is disposed between the optional protein constituent and the analyte modulating constituent. The adhesion promoter constituent can be made from any one of a wide variety of materials known in the art to facilitate the bonding between such constituents and can be applied by any one of a wide variety of methods known in the art. Typically, the adhesion promoter constituent comprises a silane compound such as γ-aminopropyltrimethoxysilane.

[0060] The use of silane coupling reagents, especially those of the formula R′Si(OR).sub.3 in which R′ is typically an aliphatic group with a terminal amine and R is a lower alkyl group, to promote adhesion is known in the art (see, e.g. U.S. Pat. No. 5,212,050 which is incorporated herein by reference). For example, chemically modified electrodes in which a silane such as γ-aminopropyltriethoxysilane and glutaraldehyde were used in a step-wise process to attach and to co-crosslink bovine serum albumin (BSA) and glucose oxidase (GO.sub.X) to the electrode surface are well known in the art (see, e.g. Yao, T. Analytica Chim. Acta 1983, 148, 27-33).

[0061] In certain embodiments of the invention, the adhesion promoting constituent further comprises one or more compounds that can also be present in an adjacent constituent such as the polydimethyl siloxane (PDMS) compounds that serves to limit the diffusion of analytes such as glucose through the analyte modulating constituent. In illustrative embodiments the formulation comprises 0.5-20% PDMS, typically 5-15% PDMS, and most typically 10% PDMS. In certain embodiments of the invention, the adhesion promoting constituent is crosslinked within the layered sensor system and correspondingly includes an agent selected for its ability to crosslink a moiety present in a proximal constituent such as the analyte modulating constituent. In illustrative embodiments of the invention, the adhesion promoting constituent includes an agent selected for its ability to crosslink an amine or carboxyl moiety of a protein present in a proximal constituent such as the analyte sensing constituent and/or the protein constituent and/or a siloxane moiety present in a compound disposed in a proximal layer such as the analyte modulating layer.

Analyte Modulating Constituent

[0062] The electrochemical sensors of the invention include an analyte modulating constituent disposed on the sensor (see, e.g. element **112** in FIG. **2**A). The term “analyte modulating constituent” is used herein according to art accepted terminology and refers to a constituent that typically forms a membrane on the sensor that operates to modulate the diffusion of one or more analytes, such as glucose, through the constituent. In certain embodiments of the invention, the analyte modulating constituent is an analyte-limiting membrane (e.g. a glucose limiting membrane) which operates to prevent or restrict the diffusion of one or more analytes, such as glucose, through the constituents. In other embodiments of the invention, the analyte-modulating constituent operates to facilitate the diffusion of one or more analytes, through the constituents. Optionally such analyte modulating constituents can be formed to prevent or restrict the diffusion of one type of molecule through the constituent (e.g. glucose), while at the same time allowing or even facilitating the diffusion of other types of molecules through the constituent (e.g. **02**).

[0063] With respect to glucose sensors, in known enzyme electrodes, glucose and oxygen from blood, as well as some interferents, such as ascorbic acid and uric acid, diffuse through a primary membrane of the sensor. As the glucose, oxygen and interferents reach the analyte sensing constituent, an enzyme, such as glucose oxidase, catalyzes the conversion of glucose to hydrogen peroxide and gluconolactone. The hydrogen peroxide may diffuse back through the analyte modulating constituent, or it may diffuse to an electrode where it can be reacted to form oxygen and a proton to produce a current that is proportional to the glucose concentration. The sensor membrane assembly serves several functions, including selectively allowing the passage of glucose therethrough. In this context, an illustrative analyte modulating constituent is a semi-permeable membrane which permits passage of water, oxygen and at least one selective analyte and which has the ability to absorb water, the membrane having a water soluble, hydrophilic polymer.

[0064] A variety of illustrative analyte modulating compositions are known in the art and are described for example in U.S. Pat. Nos. 6,319,540, 5,882,494, 5,786,439 5,777,060, 5,771,868 and

5,391,250, the disclosures of each being incorporated herein by reference. The hydrogels described therein are particularly useful with a variety of implantable devices for which it is advantageous to provide a surrounding water constituent. In some embodiments of the invention, the analyte modulating composition includes PDMS. In certain embodiments of the invention, the analyte modulating constituent includes an agent selected for its ability to crosslink a siloxane moiety present in a proximal constituent. In closely related embodiments of the invention, the adhesion promoting constituent includes an agent selected for its ability to crosslink an amine or carboxyl moiety of a protein present in a proximal constituent.

Cover Constituent

[0065] The electrochemical sensors of the invention include one or more cover constituents which are typically electrically insulating protective constituents (see, e.g. element **106** in FIG. **2**A). Typically, such cover constituents can be in the form of a coating, sheath or tube and are disposed on at least a portion of the analyte modulating constituent. Acceptable polymer coatings for use as the insulating protective cover constituent can include, but are not limited to, non-toxic biocompatible polymers such as silicone compounds, polyimides, biocompatible solder masks, epoxy acrylate copolymers, or the like. Further, these coatings can be photo-imageable to facilitate photolithographic forming of apertures through to the conductive constituent. A typical cover constituent comprises spun on silicone. As is known in the art, this constituent can be a commercially available RTV (room temperature vulcanized) silicone composition. A typical chemistry in this context is polydimethyl siloxane (acetoxy based).

C. Typical Analyte Sensor System Embodiments of the Invention

[0066] Embodiments of the sensor elements and sensors disclosed herein can be operatively coupled to a variety of other systems elements typically used with analyte sensors (e.g. structural elements such as piercing members, insertion sets and the like as well as electronic components such as processors, monitors, medication infusion pumps and the like), for example to adapt them for use in various contexts (e.g. implantation within a mammal). One embodiment of the invention includes a method of monitoring a physiological characteristic of a user using an embodiment of the invention that includes an input element capable of receiving a signal from a sensor that is based on a sensed physiological characteristic value of the user, and a processor for analyzing the received signal. In typical embodiments of the invention, the processor determines a dynamic behavior of the physiological characteristic value and provides an observable indicator based upon the dynamic behavior of the physiological characteristic value so determined. In some embodiments, the physiological characteristic value is a measure of the concentration of blood glucose in the user. In other embodiments, the process of analyzing the received signal and determining a dynamic behavior includes repeatedly measuring the physiological characteristic value to obtain a series of physiological characteristic values in order to, for example, incorporate comparative redundancies into a sensor apparatus in a manner designed to provide confirmatory information on sensor function, analyte concentration measurements, the presence of interferences and the like.

[0067] Embodiments of the invention include devices which display data from measurements of a sensed physiological characteristic (e.g. blood glucose concentrations) in a manner and format tailored to allow a user of the device to easily monitor and, if necessary, modulate the physiological status of that characteristic (e.g. modulation of blood glucose concentrations via insulin administration). An illustrative embodiment of the invention is a device comprising a sensor input capable of receiving a signal from a sensor, the signal being based on a sensed physiological characteristic value of a user; a memory for storing a plurality of measurements of the sensed physiological characteristic value of the user from the received signal from the sensor; and a display for presenting a text and/or graphical representation of the plurality of measurements of the sensed physiological characteristic value (e.g. text, a line graph or the like, a bar graph or the like, a grid pattern or the like or a combination thereof). Typically, the graphical representation displays real time measurements of the sensed physiological characteristic value. Such devices can be used in a variety of contexts, for example in combination with other medical apparatuses. In some embodiments of the invention, the device is used in combination with at least one other medical device (e.g. a glucose sensor).

[0068] An illustrative system embodiment consists of a glucose sensor, a transmitter and pump receiver and a glucose meter. In this system, radio signals from the transmitter can be sent to the pump receiver periodically (e.g. every 5 minutes) to provide real-time sensor glucose (SG) values. Values/graphs are displayed on a monitor of the pump receiver so that a user can self monitor blood glucose and deliver insulin using their own insulin pump. Typically, an embodiment of device disclosed herein communicates with a second medical device via a wired or wireless connection. Wireless communication can include, for example, the reception of emitted radiation signals as occurs with the transmission of signals via RF telemetry, infrared transmissions, optical transmission, sonic and ultrasonic transmissions and the like. Optionally, the device is an integral part of a medication infusion pump (e.g. an insulin pump). Typically in such devices, the physiological characteristic values includes a plurality of measurements of blood glucose.

D. Embodiments of the Invention and Associated Characteristics

[0069] In individuals using analyte sensors in non-hospital settings (e.g. diabetics using glucose sensors to manage their disease), relatively long sensor initialization and/or start-up periods are also problematical due to both the inconvenience to the user as well as the delayed receipt of information relating to user health. The use of glucose sensors, insulin infusion pumps and the like in the management of diabetes has increased in recent years due for example to studies showing that the morbidity and mortality issues associated with this chronic disease decrease dramatically when a patient administers insulin in a manner that closely matches the rise and fall of physiological insulin concentrations in healthy individuals. Consequently, patients who suffer from chronic diseases such as diabetes are instructed by medical personnel to play an active role in the management of their disease, in particular, the close monitoring and modulation of blood glucose levels. In this context, because many diabetics do not have medical training, they may forgo optimal monitoring and modulation of blood glucose levels due to complexities associated with such management, for example, a two hour start-up period which can be an inconvenience in view of a patient's active daily routine. For these reasons, sensors and sensor systems that are designed to include elements and/or configurations of elements can reduce sensor initialization and/or start-up times (e.g. the hydrophilic dexamethasone rejection membranes disclosed herein) are highly desirable in situations where such sensors are operated by a diabetic patient without medical training because they facilitate the patient's convenient management of their disease, behavior which is shown to decrease the well known morbidity and mortality issues observed in individuals suffering from chronic diabetes.

[0070] While the analyte sensor and sensor systems disclosed herein are typically designed to be implantable within the body of a mammal, the inventions disclosed herein are not limited to any particular environment and can instead be used in a wide variety of contexts, for example for the analysis of most in vivo and in vitro liquid samples including biological fluids such as interstitial fluids, whole-blood, lymph, plasma, serum, saliva, urine, stool, perspiration, mucus, tears, cerebrospinal fluid, nasal secretion, cervical or vaginal secretion, semen, pleural fluid, amniotic fluid, peritoneal fluid, middle ear fluid, joint fluid, gastric aspirate or the like. In addition, solid or desiccated samples may be dissolved in an appropriate solvent to provide a liquid mixture suitable for analysis.

[0071] The invention disclosed herein has a number of embodiments. One illustrative embodiment of the invention is an analyte sensor apparatus comprising: an elongated (i.e. having notably more length than width) base layer; a conductive layer disposed on the base layer and comprising a reference electrode, a working electrode and a counter electrode; an dexamethasone rejection membrane disposed upon the conductive layer, an analyte sensing layer disposed on the dexamethasone rejection membrane; an analyte modulating layer disposed on the analyte sensing layer, wherein the analyte modulating layer comprises a composition that modulates the diffusion of an analyte diffusing through the analyte modulating layer; and a cover layer disposed on the analyte sensor apparatus, wherein the cover layer comprises an aperture positioned on the cover layer so as to facilitate an analyte accessing and diffusing through the analyte modulating layer and accessing the analyte sensing layer. Typical embodiments of the invention are comprised of biocompatible materials and/or have structural features designed for implantation within a mammal. Methodological embodiments of the invention include methods for making and using the sensor embodiments disclosed herein. Certain embodiments of the invention include methods of using a specific sensor element and/or a specific constellation of sensor elements to produce and/or facilitate one or more functions of the sensor embodiments disclosed herein.

[0072] As disclosed herein, those of skill in the art understand that a conductive layer disposed on the base layer and comprising a working electrode, a counter electrode and a reference electrode includes embodiments wherein the conductive layer is disposed on at least a portion the base layer and does not necessarily completely cover the base layer. Those of skill in the art will understand that this refers to other layers within the sensor, with for example, an analyte sensing layer disposed on the conductive layer encompassing sensor embodiments where the analyte sensing layer disposed on at least a portion of the conductive layer; and an analyte modulating layer disposed on the analyte sensing layer encompassing an analyte modulating layer disposed on at least a portion of the analyte sensing layer etc. Optionally, the electrodes can be disposed on a single surface or side of the sensor structure. Alternatively, the electrodes can be disposed on a multiple surfaces or sides of the sensor structure (and can for example be connected by vias through the sensor material(s) to the surfaces on which the electrodes are disposed). In certain embodiments of the invention, the reactive surfaces of the electrodes are of different relative areas/sizes, for example a 1× reference electrode, a 2.6× working electrode and a 3.6× counter electrode.

[0073] In certain embodiments of the invention, an element of the apparatus such as an electrode or an aperture is designed to have a specific configuration and/or is made from a specific material and/or is positioned relative to the other elements so as to facilitate a function of the sensor. For example, without being bound by a specific theory or mechanism of action, it appears that sensor embodiments (e.g. simple three electrode embodiments) may be more susceptible to local environment changes around a single electrode. For example, a gas bubble on top of or close to a reference or another electrode, and/or a stagnating or semi-stagnating pool of fluid on top of or close to a reference or another electrode may consequently compromise sensor performance. In this context, a distributed electrode configuration appears be advantageous because the distribution of the electrode area allows the sensor to compensate for signal lost to a small local area (e.g. as can occur due to lack of hydration, fluid stagnation, a patient's immune response, or the like).

[0074] Typical analyte sensor apparatus embodiments comprise a plurality of working electrodes, counter electrodes and reference electrodes. Optionally, the plurality of working, counter and reference electrodes are grouped together as a unit and positionally distributed on the conductive layer in a repeating pattern of units. Alternatively, the plurality of working, counter and reference electrodes are grouped together and positionally distributed on the conductive layer in a non-repeating pattern of units. In certain embodiments of the invention, the elongated base layer is made from a material that allows the sensor to twist and bend when implanted in vivo; and the electrodes are grouped in a configuration that facilitates an in vivo fluid accessing at least one of working electrode as the sensor apparatus twists and bends when implanted in vivo. In some embodiments, the electrodes are grouped in a configuration that allows the sensor to continue to maintain an optimal function if a portion of the sensor having one or more electrodes is dislodged from an in vivo environment and exposed to an ex vivo environment.

[0075] Optionally, embodiments of the invention include a plurality of working electrodes and/or counter electrodes and/or reference electrodes (e.g. to provide redundant sensing capabilities). Such embodiments of the invention can be used in embodiments of the invention that include a processor (e.g. one linked to a program adapted for a signal subtraction/cancellation process) designed to factor out background signals in vivo, for example by comparing signal(s) at GOx coated working electrode with signal at working electrode not coated with GOx (e.g. background detection followed by a signal subtraction/cancellation process to arrive at a true signal). Certain of these embodiments of the invention are particularly useful for sensing glucose at the upper and lower ends of the glucose signal curves. Similar embodiments of the invention are used to factor out interference, for example by comparing signal(s) at GOx coated working electrode with signal at working electrode not coated with GOx. Embodiments of the invention can include a coating of a Prussian blue composition on an electrode at a location and in an amount sufficient to mediate an electrical potential of an electrode of the apparatus. Related embodiments of the invention include methods for mediating an electrical potential of an electrode of the disclosed sensor apparatus (e.g. by using a Prussian blue composition). Prussian Blue formulas are known in the art and include Fe4[Fe(CN6]3xH2O, CI no. 77510 and KFe[Fe(Cn)6]xH2O id CI no. 77520.

[0076] In typical embodiments of the invention, the sensor is operatively coupled to further elements (e.g. electronic components) such as elements designed to transmit and/or receive a signal, monitors, pumps, processors and the like. For example, in some embodiments of the invention, the sensor is operatively coupled to a sensor input capable of receiving a signal from the sensor that is based on a sensed physiological characteristic value in the mammal; and a processor coupled to the sensor input, wherein the processor is capable of characterizing one or more signals received from the sensor. A wide variety of sensor configurations as disclosed herein can be used in such systems. Optionally, for example, the sensor comprises three working electrodes, one counter electrode and one reference electrode. In certain embodiments, at least one working electrode is coated with an analyte sensing layer comprising glucose oxidase (and optionally two are coated with GOx) and at least one working electrode is not coated with an analyte sensing layer comprising glucose oxidase. Such embodiments of the invention can be used for example in sensor embodiments designed factor out background signals in vivo, for example by comparing signal(s) at GOx coated working electrode(s) with signal at working electrode(s) not coated with GOx (e.g. background detection followed by a signal subtraction/cancellation process to arrive at a true signal).

[0077] In some embodiments of sensors insertion set apparatuses, a first and a second (and/or third etc.) electrochemical sensor comprises one working, counter and reference electrode. Alternatively, the plurality of electrochemical sensors comprise a plurality of working, counter and reference electrodes, for example those having a distributed configuration as disclosed in U.S. patent application Ser. No. 11/633,254, the contents of which are incorporated by reference. In certain embodiments of the invention, at least two in the plurality of sensors are designed to measure a signal generated by the same physiological characteristic, for example blood glucose concentration. Embodiments of the invention can include, for example, a plurality of electrochemical sensors having a working electrode coated with an oxidoreductase such as glucose oxidase and are used in methods designed to sample and compare glucose concentrations observed at the plurality of in vivo insertion sites. Alternatively, at least two in the plurality of sensors in the sensor apparatus are designed to measure signals generated by the different characteristics, for example a first characteristic comprising a background or interfering signal that is unrelated to blood glucose (e.g. “interferent noise”) and a second characteristic comprising blood glucose concentrations. In an illustrative embodiment of this invention, a first sensor is designed to measure glucose oxidase and comprises one or more working electrodes coated with glucose oxidase while a second comparative sensor is designed to measure a background or interfering signal that is unrelated to blood glucose has no working electrode (or electrodes) coated with glucose oxidase.

[0078] In certain embodiments of the invention, sensor systems that utilize voltage pulsing and/or switching as disclosed herein are used in methods designed to overcome problems that can occur with implantable sensors and sensor systems due to lack of hydration (e.g. slow start-up initialization times) and/or fluid stagnation by enhancing the ability of a fluid to flow around the implanted components in a manner that inhibits the likelihood of a gas bubble or a stagnating pool of fluid from forming and/or remaining on top of or close to an electrode in a manner that compromises sensor function. In addition, embodiments of the invention that utilize voltage pulsing and/or switching can be combined with certain complementary elements disclosed herein so as to further overcome problems that result from a lack of hydration, fluid stagnation, a patient's immune response, or the like (e.g. distributed electrode configurations, multiple electrode sensors, multiple sensor apparatuses having multiple implantation sites, etc.).

[0079] In some embodiments of the invention, a processor is capable of comparing a first signal received from a working electrode in response to a first working potential with a second signal received from a working electrode in response to a second working potential, wherein the comparison of the first and second signals at the first and second working potentials can be used to identify a signal generated by an interfering compound. In one such embodiment of the invention, one working electrode is coated with glucose oxidase and another is not, and the interfering compound is acetaminophen, ascorbic acid, bilirubin, cholesterol, creatinine, dopamine, ephedrine, ibuprofen, L-dopa, methyldopa, salicylate, tetracycline, tolazamide, tolbutamide, triglycerides or uric acid. Optionally, a pulsed and/or varied (e.g. switched) voltage is used to obtain a signal from a working electrode. Typically, at least one voltage is 280, 535 or 635 millivolts. Related embodiments of the invention include methods for identifying and/or characterizing one or more signals generated by an interfering compound in various sensor embodiments of the invention (e.g. by comparing the signal from an electrode coated with an analyte sensing compound with a comparative electrode not coated with an analyte sensing compound). Optionally, such methods use a pulsed and/or varied working potential to observe a signal at an electrode.

[0080] Sensors of the invention can also be incorporated in a wide variety of medical systems known in the art. Sensors of the invention can be used, for example, in a closed loop infusion system designed to control the rate that medication is infused into the body of a user. Such a closed loop infusion system can include a sensor and an associated meter which generates an input to a controller which in turn operates a delivery system (e.g. one that calculates a dose to be delivered by a medication infusion pump). In such contexts, the meter associated with the sensor may also transmit commands to, and be used to remotely control, the delivery system. Typically, the sensor is a subcutaneous sensor in contact with interstitial fluid to monitor the glucose concentration in the body of the user, and the liquid infused by the delivery system into the body of the user includes insulin. Illustrative systems are disclosed for example in U.S. Pat. Nos. 6,558,351 and 6,551,276; PCT Application Nos. US99/21703 and US99/22993; as well as WO 2004/008956 and WO 2004/009161, all of which are incorporated herein by reference.

[0081] Certain embodiments of the invention measure peroxide and have the advantageous characteristic of being suited for implantation in a variety of sites in the mammal including regions of subcutaneous implantation and intravenous implantation as well as implantation into a variety of non-vascular regions. A peroxide sensor design that allows implantation into non-vascular regions has advantages over certain sensor apparatus designs that measure oxygen due to the problems with oxygen noise that can occur in oxygen sensors implanted into non-vascular regions. For example, in such implanted oxygen sensor apparatus designs, oxygen noise at the reference sensor can compromise the signal to noise ratio which consequently perturbs their ability to obtain stable glucose readings in this environment. The sensors of the invention therefore overcome the difficulties observed with such oxygen sensors in non-vascular regions.

[0082] In some embodiments of the invention, the analyte sensor apparatus is designed to function via anodic polarization such that the alteration in current is detected at the anodic working electrode in the conductive layer of the analyte sensor apparatus. Structural design features that can be associated with anodic polarization include designing an appropriate sensor configuration comprising a working electrode which is an anode, a counter electrode which is a cathode and a reference electrode, and then selectively disposing the appropriate analyte sensing layer on the appropriate portion of the surface of the anode within this design configuration. Optionally this anodic polarization structural design includes anodes, cathodes and/or working electrodes having different sized surface areas. For example, this structural design includes features where the working electrode (anode) and/or the coated surface of the working electrode is larger or smaller than the counter electrode (cathode) and/or the coated surface of the counter electrode (e.g. a sensor designed to have a 1× area for a reference electrode, a 2.6× area for a working electrode and a 3.6× area for a counter electrode). In this context, the alteration in current that can be detected at the anodic working electrode is then correlated with the concentration of the analyte. In certain illustrative examples of this embodiment of the invention, the working electrode is measuring and utilizing hydrogen peroxide in the oxidation reaction (see e.g. FIG. **1**), hydrogen peroxide that is produced by an enzyme such as glucose oxidase or lactate oxidase upon reaction with glucose or lactate respectively.

II. Illustrative Methods and Materials for Making Analyte Sensor Apparatus of the Invention

[0083] A number of articles, U.S. patents and patent application describe the state of the art with the common methods and materials disclosed herein and further describe various elements (and methods for their manufacture) that can be used in the sensor designs disclosed herein. These include for example, U.S. Pat. Nos. 6,413,393; 6,368,274; 5,786,439; 5,777,060; 5,391,250; 5,390,671; 5,165,407, 4,890,620, 5,390,671, 5,390,691, 5,391,250, 5,482,473, 5,299,571, 5,568,806; United States Patent Application 20020090738; as well as PCT International Publication Numbers WO 01/58348, WO 03/034902, WO 03/035117, WO 03/035891, WO 03/023388, WO 03/022128, WO 03/022352, WO 03/023708, WO 03/036255, WO03/036310 and WO 03/074107, the contents of each of which are incorporated herein by reference.

[0084] Typical sensors for monitoring glucose concentration of diabetics are further described in Shichiri, et al.: "In Vivo Characteristics of Needle-Type Glucose Sensor-Measurements of Subcutaneous Glucose Concentrations in Human Volunteers," Horm. Metab. Res., Suppl. Ser. 20:17-20 (1988); Bruckel, et al.: "In Vivo Measurement of Subcutaneous Glucose Concentrations with an Enzymatic Glucose Sensor and a Wick Method," Klin. Wochenschr. 67:491-495 (1989); and Pickup, et al.: "In Vivo Molecular Sensing in Diabetes Mellitus: An Implantable Glucose Sensor with Direct Electron Transfer," Diabetologia 32:213-217 (1989). Other sensors are described in, for example Reach, et al., in ADVANCES IN IMPLANTABLE DEVICES, A. Turner (ed.), JAI Press, London, Chap. 1, (1993), incorporated herein by reference.

A. General Methods for Making Analyte Sensors

[0085] A typical embodiment of the invention disclosed herein is a method of making a sensor apparatus for implantation within a mammal comprising the steps of: providing a base layer; forming a conductive layer on the base layer, wherein the conductive layer includes an electrode (and typically a working electrode, a reference electrode and a counter electrode); forming an dexamethasone rejection membrane on the conductive layer, forming an analyte sensing layer on the dexamethasone rejection membrane, wherein the analyte sensing layer includes a composition that can alter the electrical current at the electrode in the conductive layer in the presence of an analyte; optionally forming a protein layer on the analyte sensing layer; forming an adhesion promoting layer on the analyte sensing layer or the optional protein layer; forming an analyte modulating layer disposed on the adhesion promoting layer, wherein the analyte modulating layer includes a composition that modulates the diffusion of the analyte therethrough; and forming a cover layer disposed on at least a portion of the analyte modulating layer, wherein the cover layer further includes an aperture over at least a portion of the analyte modulating layer. In certain embodiments of the invention, the analyte modulating layer comprises a hydrophilic comb-copolymer having a central chain and a plurality of side chains coupled to the central chain, wherein at least one side chain comprises a silicone moiety. In some embodiments of these

methods, the analyte sensor apparatus is formed in a planar geometric configuration.
[0086] As disclosed herein, the various layers of the sensor can be manufactured to exhibit a variety of different characteristics which can be manipulated according to the specific design of the sensor. For example, the adhesion promoting layer includes a compound selected for its ability to stabilize the overall sensor structure, typically a silane composition. In some embodiments of the invention, the analyte sensing layer is formed by a spin coating process and is of a thickness selected from the group consisting of less than 1, 0.5, 0.25 and 0.1 microns in height. In other embodiments of the invention, the analyte sensing layer is formed by a slot coating process and is of a thickness between to 3.5 to 10 microns in height.
[0087] Typically, a method of making the sensor includes the step of forming a protein layer on the analyte sensing layer, wherein a protein within the protein layer is an albumin selected from the group consisting of bovine serum albumin and human serum albumin. Typically, a method of making the sensor includes the step of forming an analyte sensing layer that comprises an enzyme composition selected from the group consisting of glucose oxidase, glucose dehydrogenase, lactate oxidase, hexokinase and lactate dehydrogenase. In such methods, the analyte sensing layer typically comprises a carrier protein composition in a substantially fixed ratio with the enzyme, and the enzyme and the carrier protein are distributed in a substantially uniform manner throughout the analyte sensing layer.
B. Typical Protocols and Materials Useful in the Manufacture of Analyte Sensors
[0088] The disclosure provided herein includes sensors and sensor designs that can be generated using combinations of various well known techniques. The disclosure further provides methods for applying very thin enzyme coatings to these types of sensors as well as sensors produced by such processes. In this context, some embodiments of the invention include methods for making such sensors on a substrate according to art accepted processes. In certain embodiments, the substrate comprises a rigid and flat structure suitable for use in photolithographic mask and etch processes. In this regard, the substrate typically defines an upper surface having a high degree of uniform flatness. A polished glass plate may be used to define the smooth upper surface. Alternative substrate materials include, for example, stainless steel, aluminum, and plastic materials such as delrin, etc. In other embodiments, the substrate is non-rigid and can be another layer of film or insulation that is used as a substrate, for example plastics such as polyimides and the like.
[0089] An initial step in the methods of the invention typically includes the formation of a base layer of the sensor. The base layer can be disposed on the substrate by any desired means, for example by controlled spin coating. In addition, an adhesive may be used if there is not sufficient adhesion between the substrate layer and the base layer. A base layer of insulative material is formed on the substrate, typically by applying the base layer material onto the substrate in liquid form and thereafter spinning the substrate to yield the base layer of thin, substantially uniform thickness. These steps are repeated to build up the base layer of sufficient thickness, followed by a sequence of photolithographic and/or chemical mask and etch steps to form the conductors discussed below. In an illustrative form, the base layer comprises a thin film sheet of insulative material, such as ceramic or polyimide substrate. The base layer can comprise an alumina substrate, a polyimide substrate, a glass sheet, controlled pore glass, or a planarized plastic liquid crystal polymer. The base layer may be derived from any material containing one or more of a variety of elements including, but not limited to, carbon, nitrogen, oxygen, silicon, sapphire, diamond, aluminum, copper, gallium, arsenic, lanthanum, neodymium, strontium, titanium, yttrium, or combinations thereof. Additionally, the substrate may be coated onto a solid support by a variety of methods well-known in the art including physical vapor deposition, or spin-coating with materials such as spin glasses, chalcogenides, graphite, silicon dioxide, organic synthetic polymers, and the like.
[0090] The methods of the invention further include the generation of a conductive layer having one or more sensing elements. Typically these sensing elements are electrodes that are formed by

one of the variety of methods known in the art such as photoresist, etching and rinsing to define the geometry of the active electrodes. The electrodes can then be made electrochemically active, for example by electrodeposition of Pt black for the working and counter electrode, and silver followed by silver chloride on the reference electrode. A sensor layer such as an analyte sensing enzyme layer can then be disposed on the sensing layer by electrochemical deposition or a method other than electrochemical deposition such as spin coating, followed by vapor crosslinking, for example with a dialdehyde (glutaraldehyde) or a carbodi-imide.

[0091] Electrodes of the invention can be formed from a wide variety of materials known in the art. For example, the electrode may be made of a noble late transition metal. Metals such as gold, platinum, silver, rhodium, iridium, ruthenium, palladium, or osmium can be suitable in various embodiments of the invention. Other compositions such as carbon or mercury can also be useful in certain sensor embodiments. Of these metals, silver, gold, or platinum is typically used as a reference electrode metal. A silver electrode which is subsequently chloridized is typically used as the reference electrode. These metals can be deposited by any means known in the art, including the electrodeposition method cited, supra, or by an electroless method which may involve the deposition of a metal onto a previously metallized region when the substrate is dipped into a solution containing a metal salt and a reducing agent. The electroless method proceeds as the reducing agent donates electrons to the conductive (metallized) surface with the concomitant reduction of the metal salt at the conductive surface. The result is a layer of adsorbed metal. (For additional discussions on electroless methods, see: Wise, E. M. Palladium: Recovery, Properties, and Uses, Academic Press, New York, New York (1988); Wong, K. et al. Plating and Surface Finishing 1988, 75, 70-76; Matsuoka, M. et al. Ibid. 1988, 75, 102-106; and Pearlstein, F. “Electroless Plating,” Modern Electroplating, Lowenheim, F. A., Ed., Wiley, New York, N.Y. (1974), Chapter 31). Such a metal deposition process must yield a structure with good metal to metal adhesion and minimal surface contamination, however, to provide a catalytic metal electrode surface with a high density of active sites. Such a high density of active sites is a property necessary for the efficient redox conversion of an electroactive species such as hydrogen peroxide.

[0092] In an exemplary embodiment of the invention, the base layer is initially coated with a thin film conductive layer by electrode deposition, surface sputtering, or other suitable process step. In one embodiment this conductive layer may be provided as a plurality of thin film conductive layers, such as an initial chrome-based layer suitable for chemical adhesion to a polyimide base layer followed by subsequent formation of thin film gold-based and chrome-based layers in sequence. In alternative embodiments, other electrode layer conformations or materials can be used. The conductive layer is then covered, in accordance with conventional photolithographic techniques, with a selected photoresist coating, and a contact mask can be applied over the photoresist coating for suitable photoimaging. The contact mask typically includes one or more conductor trace patterns for appropriate exposure of the photoresist coating, followed by an etch step resulting in a plurality of conductive sensor traces remaining on the base layer. In an illustrative sensor construction designed for use as a subcutaneous glucose sensor, each sensor trace can include three parallel sensor elements corresponding with three separate electrodes such as a working electrode, a counter electrode and a reference electrode.

[0093] Portions of the conductive sensor layers are typically covered by an insulative cover layer, typically of a material such as a silicon polymer and/or a polyimide. The insulative cover layer can be applied in any desired manner. In an exemplary procedure, the insulative cover layer is applied in a liquid layer over the sensor traces, after which the substrate is spun to distribute the liquid material as a thin film overlying the sensor traces and extending beyond the marginal edges of the sensor traces in sealed contact with the base layer. This liquid material can then be subjected to one or more suitable radiation and/or chemical and/or heat curing steps as are known in the art. In alternative embodiments, the liquid material can be applied using spray techniques or any other desired means of application. Various insulative layer materials may be used such as photoimagable epoxyacrylate, with an illustrative material comprising a photoimagable polyimide available from OCG, Inc. of West Paterson, N.J., under the product number 7020.

[0094] In certain embodiments of the invention materials used to form one or more layers of a sensor stack are selected to control their diffusion coefficients for one or more compounds such as O.sub.2 or glucose. Typically, for example, materials forming the dexamethasone rejection membrane and/or materials forming the analyte modulating layer are selected so that the diffusivity of O.sub.2 diffusion coefficient through said layers is at least 1.0×10.sup.−5 cm.sup.2/s 37° C. in phosphate buffered saline (e.g. is between 1.0 and 3.0×10.sup.−5 cm.sup.2/s 37° C.). Similarly, in illustrative embodiments of the invention, the materials forming the dexamethasone rejection membrane and/or materials forming the analyte modulating layer are selected so as to exhibit a glucose permeability of at least 1×10.sup.−8 cm.sup.2/s at 37° C. in phosphate buffered saline.

[0095] In an illustrative sensor embodiment for use as a glucose sensor, an enzyme (typically glucose oxidase) is coated with the enzyme so as to define a working electrode. One or both of the other electrodes can be provided with the same coating as the working electrode. Alternatively, the other two electrodes can be provided with other suitable chemistries, such as other enzymes, left uncoated, or provided with chemistries to define a reference electrode and a counter electrode for the electrochemical sensor. Methods for producing the enzyme coatings include spin coating processes, dip and dry processes, low shear spraying processes, ink-jet printing processes, silk screen processes and the like. Typically, such coatings are vapor crosslinked subsequent to their application. Surprisingly, sensors produced by these processes have material properties that exceed those of sensors having coatings produced by electrodeposition including enhanced longevity, linearity, regularity as well as improved signal to noise ratios. In addition, embodiments of the invention that utilize glucose oxidase coatings formed by such processes are designed to recycle hydrogen peroxide and improve the biocompatibility profiles of such sensors.

[0096] Sensors generated by processes such as spin coating processes also avoid other problems associated with electrodeposition, such as those pertaining to the material stresses placed on the sensor during the electrodeposition process. In particular, the process of electrodeposition is observed to produce mechanical stresses on the sensor, for example mechanical stresses that result from tensile and/or compression forces. In certain contexts, such mechanical stresses may result in sensors having coatings with some tendency to crack or delaminate. This is not observed in coatings disposed on sensor via spin coating or other low-stress processes.

[0097] In some embodiments of the invention, the sensor is made by methods which apply an analyte modulating layer that comprises a hydrophilic membrane coating which can regulate the amount of analyte that can access the enzyme of the sensor layer. For example, the cover layer that is added to the glucose sensors of the invention can comprise a glucose limiting membrane, which regulates the amount of glucose that access glucose oxidase enzyme layer on an electrode. Such glucose limiting membranes can be made from a wide variety of materials known to be suitable for such purposes, e.g., silicones such as polydimethyl siloxane and the like, polyurethanes, cellulose acetates, NAFION, polyester sulfonic acids (e.g. Kodak AQ), hydrogels or any other membrane known to those skilled in the art that is suitable for such purposes. In certain embodiments of the invention, the analyte modulating layer comprises a hydrophilic comb-copolymer having a central chain and a plurality of side chains coupled to the central chain, wherein at least one side chain comprises a silicone moiety. In some embodiments of the invention pertaining to sensors having hydrogen peroxide recycling capabilities, the membrane layer that is disposed on the glucose oxidase enzyme layer functions to inhibit the release of hydrogen peroxide into the environment in which the sensor is placed and to facilitate hydrogen peroxide molecules and accessing the electrode sensing elements.

[0098] In some embodiments of the methods of invention, an adhesion promoter layer is disposed between a cover layer (e.g. an analyte modulating membrane layer) and a analyte sensing layer in order to facilitate their contact and is selected for its ability to increase the stability of the sensor apparatus. As noted herein, compositions of the adhesion promoter layer are selected to provide a number of desirable characteristics in addition to an ability to provide sensor stability. For example, some compositions for use in the adhesion promoter layer are selected to play a role in dexamethasone rejection as well as to control mass transfer of the desired analyte. The adhesion promoter layer can be made from any one of a wide variety of materials known in the art to facilitate the bonding between such layers and can be applied by any one of a wide variety of methods known in the art. Typically, the adhesion promoter layer comprises a silane compound such as γ-aminopropyltrimethoxysilane. In certain embodiments of the invention, the adhesion promoting layer and/or the analyte modulating layer comprises an agent selected for its ability to crosslink a siloxane moiety present in a proximal layer. In other embodiments of the invention, the adhesion promoting layer and/or the analyte modulating layer comprises an agent selected for its ability to crosslink an amine or carboxyl moiety of a protein present in a proximal layer. In an optional embodiment, the AP layer further comprises Polydimethyl Siloxane (PDMS), a polymer typically present in analyte modulating layers such as a glucose limiting membrane. In illustrative embodiments the formulation comprises 0.5-20% PDMS, typically 5-15% PDMS, and most typically 10% PDMS. The addition of PDMS to the AP layer can be advantageous in contexts where it diminishes the possibility of holes or gaps occurring in the AP layer as the sensor is manufactured.

[0099] One illustrative embodiment of the invention is a method of making a sensor electrode by providing an electroactive surface which can function as an electrode (e.g. platinum), forming an dexamethasone rejection membrane on the electroactive surface, spin coating, slot coating, or spray coating an enzyme layer on the DRM and then forming an analyte contacting layer (e.g. an analyte modulating layer such as a glucose limiting membrane) on the electrode, wherein the analyte contacting layer regulates the amount of analyte that can contact the enzyme layer. In some methods, the enzyme layer is vapor crosslinked on the sensor layer. In a typical embodiment of the invention, a sensor is formed to include at least one working electrode and at least one counter electrode. In certain embodiments, the DRM is formed on at least a portion of the working electrode and at least a portion of the counter electrode. Typically, the enzyme layer comprises one or more enzymes such as glucose oxidase, glucose dehydrogenase, lactate oxidase, hexokinase or lactate dehydrogenase and/or like enzymes. In a specific method, the enzyme layer comprises glucose oxidase that is stabilized by coating it on the sensor layer in combination with a carrier protein in a fixed ratio. Typically the carrier protein is albumin. Typically such methods include the step of forming an adhesion promoter layer disposed between the glucose oxidase layer and the analyte contacting layer. Optionally, a layer such as the DRM and/or the adhesion promoter layer is subjected to a curing process prior to the formation of the analyte contacting layer.

[0100] The finished sensors produced by such processes are typically quickly and easily removed from a supporting substrate (if one is used), for example, by cutting along a line surrounding each sensor on the substrate. The cutting step can use methods typically used in this art such as those that include a UV laser cutting device that is used to cut through the base and cover layers and the functional coating layers along a line surrounding or circumscribing each sensor, typically in at least slight outward spaced relation from the conductive elements so that the sufficient interconnected base and cover layer material remains to seal the side edges of the finished sensor. In addition, dicing techniques typically used to cut ceramic substrates can be used with the appropriate sensor embodiments. Since the base layer is typically not physically attached or only minimally adhered directly to the underlying supporting substrate, the sensors can be lifted quickly and easily from the supporting substrate, without significant further processing steps or potential damage due to stresses incurred by physically pulling or peeling attached sensors from the supporting substrate. The supporting substrate can thereafter be cleaned and reused, or otherwise discarded. The functional coating layer(s) can be applied either before or after other sensor components are removed from the supporting substrate (e.g., by cutting).

III. Methods for Using Analyte Sensor Apparatus of the Invention
[0101] Related embodiments of the invention is a method of sensing an analyte within the body of a mammal, the method comprising implanting an analyte sensor embodiment disclosed herein in to the mammal and then sensing one or more electrical fluctuations such as alteration in current at the working electrode and correlating the alteration in current with the presence of the analyte, so that the analyte is sensed. In one such method, the analyte sensor apparatus senses glucose in the mammal. In an alternative method, the analyte sensor apparatus senses lactate, potassium, calcium, oxygen, pH, and/or any physiologically relevant analyte in the mammal.
[0102] Certain analyte sensors having the structure discussed above have a number of highly desirable characteristics which allow for a variety of methods for sensing analytes in a mammal. For example in such methods, the analyte sensor apparatus implanted in the mammal functions to sense an analyte within the body of a mammal for more than 1, 2, 3, 4, 5, or 6 months. Typically, the analyte sensor apparatus so implanted in the mammal senses an alteration in current in response to an analyte within 15, 10, 5 or 2 minutes of the analyte contacting the sensor. In such methods, the sensors can be implanted into a variety of locations within the body of the mammal, for example in both vascular and non-vascular spaces.
IV. Kits and Sensor Sets of the Invention
[0103] In another embodiment of the invention, a kit and/or sensor set, useful for the sensing an analyte as is described above, is provided. The kit and/or sensor set typically comprises a container, a label and an analyte sensor as described above. Suitable containers include, for example, an easy to open package made from a material such as a metal foil, bottles, vials, syringes, and test tubes. The containers may be formed from a variety of materials such as metals (e.g. foils) paper products, glass or plastic. The label on, or associated with, the container indicates that the sensor is used for assaying the analyte of choice. In some embodiments, the container holds a porous matrix that is coated with a layer of an enzyme such as glucose oxidase. The kit and/or sensor set may further include other materials desirable from a commercial and user standpoint, including elements or devices designed to facilitate the introduction of the sensor into the analyte environment, other buffers, diluents, filters, needles, syringes, and package inserts with instructions for use.
CONCLUSION
[0104] This concludes the description of the preferred embodiment of the present invention. The foregoing description of one or more embodiments of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching.
[0105] All publications, patents, and patent applications cited herein are hereby incorporated by reference in their entirety for all purposes.

Claims

1. A method of making a sensor apparatus for implantation within a mammal comprising the steps of: providing a base layer; forming a conductive layer on the base layer, wherein the conductive layer includes a working electrode; forming a dexamethasone rejection membrane; forming an analyte sensing layer, wherein the analyte sensing layer includes an oxidoreductase; and forming an analyte modulating layer, wherein the analyte modulating layer includes a composition that modulates the diffusion of the analyte therethrough.
2. The method of claim 1, wherein the method includes disposing a dexamethasone compound on the sensor.
3. The method of claim 1, wherein the dexamethasone rejection membrane comprises a Poly(2-hydroxyethyl methacrylate) composition.

4. The method of claim 3, wherein the Poly(2-hydroxyethyl methacrylate) composition comprises Poly(2-hydroxyethyl methacrylate) polymers.
5. The method of claim 1, wherein the Poly(2-hydroxyethyl methacrylate) composition comprises prepolymerized Poly(2-hydroxyethyl methacrylate) monomers.
6. The method of claim 1, wherein the dexamethasone rejection membrane comprises about: 50%-80% (w/v) tetra (ethylene glycol) diacrylate; in combination with 0.5%-10% (w/v) methacryloyloxyethyl phosphorylcholine (MPC).
7. The method of claim 1, wherein the dexamethasone rejection membrane comprises about: 60%-70% (w/v) tetra (ethylene glycol) diacrylate; in combination with 1.0%-1% (w/v) methacryloyloxyethyl phosphorylcholine (MPC).
8. A sensor apparatus for implantation within a mammal comprising: a base layer; a conductive layer disposed on the base layer, wherein the conductive layer includes a working electrode; a dexamethasone rejection membrane disposed over the working electrode; an analyte sensing layer, wherein the analyte sensing layer includes an oxidoreductase; and an analyte modulating layer, wherein the analyte modulating layer includes a composition that modulates the diffusion of the analyte therethrough.
9. The sensor apparatus of claim 8, further comprising a dexamethasone compound.
10. The sensor apparatus of claim 8, wherein the dexamethasone rejection membrane comprises a Poly(2-hydroxyethyl methacrylate) composition.
11. The sensor apparatus of claim 10, wherein the Poly(2-hydroxyethyl methacrylate) composition comprises Poly(2-hydroxyethyl methacrylate) polymers.
12. The sensor apparatus of claim 10, wherein the Poly(2-hydroxyethyl methacrylate) composition comprises prepolymerized Poly(2-hydroxyethyl methacrylate) monomers.
11. The sensor apparatus of claim 10, wherein the dexamethasone rejection membrane comprises about: 50%-80% (w/v) tetra (ethylene glycol) diacrylate; in combination with 0.5%-10% (w/v) methacryloyloxyethyl phosphorylcholine (MPC).
12. The sensor apparatus of claim 8, wherein the dexamethasone rejection membrane comprises about: 60%-70% (w/v) tetra (ethylene glycol) diacrylate; in combination with 1.0%-1% (w/v) methacryloyloxyethyl phosphorylcholine (MPC).
13. A method of estimating the concentrations of glucose in vivo, the method comprising: disposing a sensor apparatus of claim 1 into an in vivo environment of a subject, wherein the environment comprises glucose; and estimating the concentration of glucose; so that the concentrations of glucose vivo are estimated.
14. The method of claim 13, wherein the sensor apparatus comprises a dexamethasone compound.
15. The method of claim 13, wherein the dexamethasone rejection membrane comprises a Poly(2-hydroxyethyl methacrylate) composition.